



ACTAI • WEEK 02 • LECTURE 02

NotebookLM

Your personal AI research assistant — grounded in your own documents.

A hands-on build-along — follow along in your own notebook

THE BIG IDEA

What is NotebookLM?

A free AI tool from Google that you feed your own sources — then it answers your questions using only those sources, with citations you can click.



You add the sources

PDFs, docs, slides, websites,
YouTube, audio



It answers from them

every claim links back to where it
came from



It transforms them

podcasts, videos, study guides,
slides & more

Grounded, not guessing

A normal chatbot

- Answers from everything it ever read
- Can confidently invent facts
- No idea what's in your documents
- You can't check where it got that



NotebookLM

- Answers ONLY from your sources
- Far fewer made-up facts
- Every answer is cited — click to verify
- Your sources stay private, not used to train AI

Perfect for studying, research, and making sense of long or messy documents.

GETTING STARTED

Free, and you already have access



notebooklm.google.com

Free with any Google account — just sign in. Nothing to install.



Web + mobile apps

Works in the browser, plus iOS and Android apps.



Private by default

Your sources aren't used to train Google's models.



Generous free tier

Plenty for a course; paid Plus only for heavy use.

STEP 1

Create a notebook & add your sources



Click 'Create new', then 'Add sources'. This is the material NotebookLM will read and ground every answer in.

- PDFs, Google Docs & Slides, Word files
- Websites and YouTube video links
- Pasted text and copied notes
- Audio files (recorded lectures, voice notes)

DEMO LIVE: Create a notebook and drop in a PDF + a YouTube link, live.

How many, and finding more



Good habits

- 3–10 focused sources beat 50 random ones
- Up to 50 sources per notebook (free)
- Each source can be very long (~500k words)
- Name sources clearly so you can cite them



Discover sources

Don't have sources yet? Click 'Discover' and describe a topic — NotebookLM finds relevant articles from the web for you to add with one click.

THE LAYOUT

Three panels you'll live in



Sources

Left panel.
Everything you uploaded. Tick which sources an answer should use.



Chat

Centre.
Ask questions and get cited answers. Save the good ones as notes.



Studio

Right panel.
One-click tools that turn your sources into audio, video, guides & more.

Ask questions, get cited answers



Type any question about your sources. Every answer comes with little numbered citations — click one to jump to the exact spot in your source.

- Ask for summaries, explanations, comparisons
- Click a citation to verify it instantly
- Save a great answer as a Note (then reuse it as a source)
- Try 'Learning Guide' mode — it asks you questions to deepen understanding

DEMO LIVE: Ask 'What are the main arguments?' then click a citation to show the source.

Set up your notebook



Try it now



10 min

- Create a new notebook.
- Add 2–3 sources (a PDF, a website, or a YouTube link).
- Ask the three questions on the right.
- Click a citation to see where the answer came from.

ASK YOUR NOTEBOOK THESE

“Give me a 5-point summary.”

“What are the key terms I should know?”

“Explain the hardest part simply.”

No source handy? Use 'Discover sources' and type any topic from your course.

Turn your sources into nine things



Audio Overview



Video Overview



Mind Map



Reports



Flashcards



Quizzes



Infographics



Slide Decks



Data Tables

All one-click. All grounded in your sources. We'll try the big ones — the rest work the same way.

Instant study guides & briefings



One click turns your sources into a polished document. Pick the format you need.

- Study Guide — key concepts + practice questions
- Briefing Doc — the essentials, summarised
- FAQ — the obvious questions, answered
- Timeline — events in order
- Blog Post — a readable write-up

DEMO LIVE: Generate a Study Guide from your sources and scroll through it.

See everything at a glance



Generates an interactive map of every idea in your sources and how they connect — great before you start reading.

- Expand and collapse branches to explore
- Click any branch to open a chat about just that topic
- Shows you what's worth digging into
- Download it as an image

DEMO LIVE: Generate a Mind Map, expand a branch, then click it to chat on that topic.

Flashcards & Quizzes



Flashcards

- Auto-made from your sources
- Set how many and how hard
- Flip through to self-test



Quizzes

- Generates a test from your material
- Adjust topic and difficulty
- Quick practice before an exam

Upload your lecture notes, hit generate, and you have a revision set in seconds.

Make your study set



Try it now



10 min

- Open the Studio panel.
- Generate a Study Guide from your sources.
- Generate a set of Flashcards.
- Generate a short Quiz and try answering it.

WHAT TO NOTICE

How accurate is it to your sources?

Everything traces back to what you uploaded.

Tip: you can ask for a focus,

e.g. 'flashcards on chapter 3 only'.

Audio Overview — your AI podcast



Turns your sources into a realistic podcast where two AI hosts chat through and explain your material. Not robotic — they banter and use analogies.

- One click generates a full episode
- Customise the focus (e.g. 'just chapter 4')
- Set the level (beginner vs advanced)
- Available in 80+ languages
- Brilliant for revising on the go

DEMO LIVE: Generate an Audio Overview now (it takes a few minutes — start it, then continue).

You can talk back to it



Interactive mode

While the podcast plays, tap “Join” to interrupt and ask the hosts a question — they answer, then carry on. It's a conversation, not just a recording.



Any language

Generate the podcast in 80+ languages — great for studying in your first language, or practising another.

Video Overview — a narrated slide deck



Like the Audio Overview, but visual: an AI-narrated video that walks through your sources with slides, diagrams, and images.

- Great for visual learners
- Combines narration + visuals automatically
- Also available in many languages
- Takes a little longer to generate than audio

DEMO LIVE: Kick off a Video Overview early so one is ready to show by end of class.

Generate your Audio Overview



Try it now



12 min

- In Studio, click Audio Overview.
- Optional: set a focus and the level.
- Generate it (this takes a few minutes).
- While it builds, generate a Mind Map too.
- Listen — then try 'Join' to ask a question.

WHILE IT GENERATES

Audio takes a few minutes — that's normal.

Use the wait to explore your Mind Map and click a branch to chat on that topic.

Then play your podcast and tap Join to ask the hosts something.

Turn sources into deliverables



Slide Decks

Presentation slides you can download as PPTX.



Infographics

Visual one-pagers; can match your brand colours.



Data Tables

Pull scattered facts into a sortable table.

Download anything: Studio outputs export as PPTX, PDF, or audio — so your notebook becomes finished work you can hand in or present.

SHARE & TAKE IT WITH YOU

Collaborate and study anywhere



Share a notebook

Send a notebook to classmates to study together (they sign in with Google). Great for group projects and shared revision.



Mobile app

Get the iOS or Android app to listen to your Audio Overviews on the bus, at the gym, anywhere.

FOR EVERY FIELD

How you'll actually use it



Students

turn lecture notes into guides,
flashcards & a podcast



Research

question a stack of papers and
trust the citations



Writing

organise sources, draft, and check
every claim



Business

digest reports, contracts, and long
email threads



Teaching

build study sets and explainers
from your material



Anyone busy

listen to long documents instead
of reading them

Know the limits



It only knows your sources

Garbage in, garbage out — add good, relevant material.



Still worth a sanity check

Grounding cuts errors, but verify anything critical.



Free limits exist

e.g. a few Audio Overviews per day — plenty for class.



Fewer, focused sources win

3–10 strong sources beat dumping in everything.



LECTURE RECAP

You've got a research superpower.

Sources in, cited answers out — plus study guides, flashcards, quizzes, mind maps, and your own AI podcast and video, all grounded in your material.

Homework · build a notebook for one of your real courses. **Next** — your wider AI toolkit.